

Emrul Hasan Emon

Software Engineer | Grameen HealthTech Limited

emonemrulhasan35@gmail.com | Dhaka, Bangladesh | +880 1625 054331 | [Linked In](#) | [GitHub](#)
[Medium](#)

SKILLS

Programming Language: Java, XSL, Go, Python, JavaScript, TypeScript, C, C++

Web Development: Nest JS, GIN, Angular, PrimeNg, BullMQ, LangChain, LangGraph

Database: PostgreSQL, MySQL, Redis, MongoDB

Search Services: Open Search, Elastic Search, Endeca Search

ORM Tools: TypeORM, Mongoose, GORM

Containers & Orchestration: Docker, Kubernetes

AI & Productivity Tools: Copilot, Claude Code, ChatGPT, Gemini, Jira (Project Management)

Problem Solving: [Codeforces \[EXPERT. 2000+ Problems Solved.\]](#), [Leetcode \[150+ Problems Solved\]](#)

Language: Bangla & English

EXPERIENCE

Grameen HealthTech Limited

July, 2025 - Present

Software Engineer - Full Time

Project: Shukhee ([Try It](#))

- Contributed to database design and developed microservices for E-commerce **Admin**, **Seller**, **Customer** and **Rider** domains of the **Shukhee Healthstore platform**, including integration between the **Warehouse Management System** and **Seller Portal**..
- Built a dedicated **Search Service** by **Open Search** for Product Discovery by product name, category and attribute filtering, accelerating product discovery speed by 3x.
- Extended core **Admin** and **SSK Portal** microservices to support multi-organization workflows, simplifying multi-tenant maintenance and preventing code duplication.
- Co-developed centralized **Payment** and **Coupon** services using a shared schema which ensures uniform configurations and eliminates the need to manage separate data stores.
- Implemented a **Redis-backed Single Device Login** mechanism, leveraging in-memory caching to instantly eliminate concurrent sessions and strengthen security.
- Integrated **BullMQ** for intensive background job processing, offloading heavy computations from the main event loop to improve API responsiveness.
- Authored comprehensive microservice API documentation, managed project deliverables via **Jira** and maintained a strict development-to-production (**Dev-Stage-Prod**) release lifecycle.
- Automated financial operations by integrating **Tally** with **Shukhee** and contributed to the development of the **Shukhee AI Assistant**.
- Tech Stack:** NestJS, Python, BullMQ, Postgres, MySQL, TypeORM, MongoDB, Mongoose, NextJs

Kaz Software (Full Time)
Associate Software Engineer

October, 2023 - June, 2025

Project: Tax Research Platform (TRP) (Try It)

- Co-developed microservices and database schemas for **Search, Tax Document Processing, Export** and **Grid Comparison modules**, writing comprehensive unit and integration tests.
- Co-led the microservices migration from **Endeca** to **Elasticsearch** to boost performance and designed a Factory Pattern architecture to maintain dynamic compatibility during the transition.
- Integrated **Elasticsearch** into the frontend layer to replace legacy **Endeca** workflows, optimizing the user experience for tax document queries, document exports and so on.
- Optimized the Document Comparison feature by refactoring the frontend codebase, accelerating page loading speeds by 3x.
- Authored comprehensive microservice API documentation, managed project deliverables via **Jira** and maintained a strict development-to-production (**Dev-Stage-Prod**) release lifecycle.
- **Tech Stack:** Java, Go, XSL, MySQL, Endeca Search, Elastic Search, Angular, PrimeNg.

EDUCATION

B.Sc. in Computer Science & Engineering

International University of Business Agriculture and Technology | JAN 2020 – MAY 2024 | CGPA - 3.93 | Dhaka, Bangladesh

Co-Curricular Activities: Trained students in C, C++, and problem-solving at IUBAT; served as Deputy Manager of the Programming Society (2021–2022). Organized and judged multiple programming contests, including NGPC Girls Selection (2022 & 2023).

RESEARCH PROJECTS

Sentiment Analysis on Bengali Text Using Deep Learning

An NLP research project on Bengali sentiment analysis using **~2,000** text samples to classify sentiments as positive or negative. Applied **Bangla BERT** with appropriate text preprocessing, achieving 90%+ accuracy in sentiment classification.

ACHIEVEMENTS & PARTICIPATION

- **Champion**, Intra University Programming Contest, **Summer – 2022, IUBAT & Fall – 2021, IUBAT.**
- **46th** at **ICPC Asia West Continent Final Contest – 2023** (Team: IUBAT_Potato_MIND)
- **49th** at **ICPC Asia Dhaka Regional Contest – BUBT, 2023** (Team: IUBAT_Potato_Mind)
- **50th** at **Cefalo SUST Inter University Programming Contest 2023** (Team: IUBAT_Potato_Mind). Also participated in 8+ Inter University Programming Contests.
- **59th** at **SELISE Coding Challenge 2023** (Solo Contest. Username: emrulhasan)